

student analysis

January 11, 2026

1 Student Performance Analysis

```
[180]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1.1 Data Loading and Cleaning

```
[181]: df=pd.read_csv("student_score.csv")
print(df.head())
```

	Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	0	female	NaN	bachelor's degree	standard	none	
1	1	female	group C	some college	standard	NaN	
2	2	female	group B	master's degree	standard	none	
3	3	male	group A	associate's degree	free/reduced	none	
4	4	male	group C	some college	standard	none	

	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	\
0	married	regularly	yes	3.0	school_bus	
1	married	sometimes	yes	0.0	NaN	
2	single	sometimes	yes	4.0	school_bus	
3	married	never	no	1.0	NaN	
4	married	sometimes	yes	0.0	school_bus	

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74
1	5 - 10	69	90	88
2	< 5	87	93	91
3	5 - 10	45	56	42
4	5 - 10	76	78	75

```
[182]: df.describe()
```

```
[182]:
```

	Unnamed: 0	NrSiblings	MathScore	ReadingScore	WritingScore
count	30641.000000	29069.000000	30641.000000	30641.000000	30641.000000
mean	499.556607	2.145894	66.558402	69.377533	68.418622

std	288.747894	1.458242	15.361616	14.758952	15.443525
min	0.000000	0.000000	0.000000	10.000000	4.000000
25%	249.000000	1.000000	56.000000	59.000000	58.000000
50%	500.000000	2.000000	67.000000	70.000000	69.000000
75%	750.000000	3.000000	78.000000	80.000000	79.000000
max	999.000000	7.000000	100.000000	100.000000	100.000000

[183]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30641 entries, 0 to 30640
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            30641 non-null  int64
1   Gender                30641 non-null  object
2   EthnicGroup           28801 non-null  object
3   ParentEduc            28796 non-null  object
4   LunchType             30641 non-null  object
5   TestPrep              28811 non-null  object
6   ParentMaritalStatus   29451 non-null  object
7   PracticeSport         30010 non-null  object
8   IsFirstChild          29737 non-null  object
9   NrSiblings            29069 non-null  float64
10  TransportMeans        27507 non-null  object
11  WklyStudyHours        29686 non-null  object
12  MathScore             30641 non-null  int64
13  ReadingScore          30641 non-null  int64
14  WritingScore          30641 non-null  int64
dtypes: float64(1), int64(4), object(10)
memory usage: 3.5+ MB
```

[184]: `df.isnull().sum()`

```
[184]: Unnamed: 0            0
      Gender              0
      EthnicGroup        1840
      ParentEduc         1845
      LunchType           0
      TestPrep           1830
      ParentMaritalStatus 1190
      PracticeSport       631
      IsFirstChild        904
      NrSiblings          1572
      TransportMeans      3134
      WklyStudyHours      955
      MathScore           0
      ReadingScore        0
```

```
WritingScore          0
dtype: int64
```

```
[185]: #drop unnamed column
df=df.drop("Unnamed: 0",axis=1,errors="ignore")
print(df.head())
```

	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	female	NaN	bachelor's degree	standard	none	
1	female	group C	some college	standard	NaN	
2	female	group B	master's degree	standard	none	
3	male	group A	associate's degree	free/reduced	none	
4	male	group C	some college	standard	none	

	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	\
0	married	regularly	yes	3.0	school_bus	
1	married	sometimes	yes	0.0	NaN	
2	single	sometimes	yes	4.0	school_bus	
3	married	never	no	1.0	NaN	
4	married	sometimes	yes	0.0	school_bus	

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74
1	5 - 10	69	90	88
2	< 5	87	93	91
3	5 - 10	45	56	42
4	5 - 10	76	78	75

```
[186]: #change weekly study hours
df["WklyStudyHours"]=df["WklyStudyHours"].str.replace("05-Oct","5-10")
df.head()
```

```
[186]:
```

	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	female	NaN	bachelor's degree	standard	none	
1	female	group C	some college	standard	NaN	
2	female	group B	master's degree	standard	none	
3	male	group A	associate's degree	free/reduced	none	
4	male	group C	some college	standard	none	

	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	\
0	married	regularly	yes	3.0	school_bus	
1	married	sometimes	yes	0.0	NaN	
2	single	sometimes	yes	4.0	school_bus	
3	married	never	no	1.0	NaN	
4	married	sometimes	yes	0.0	school_bus	

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74

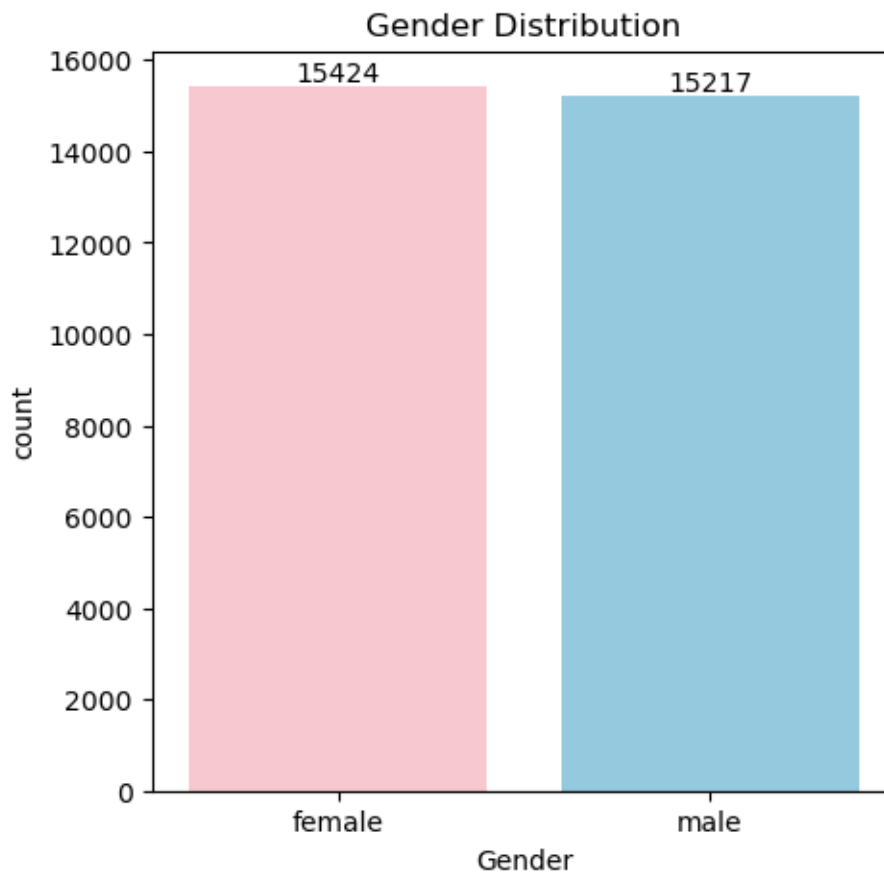
1	5 - 10	69	90	88
2	< 5	87	93	91
3	5 - 10	45	56	42
4	5 - 10	76	78	75

1.2 Gender Distribution

```
[187]: #gender distribution
plt.figure(figsize=(5,5))
plt.title("Gender Distribution")
ax = sns.countplot(
    data=df,
    x="Gender",
    hue="Gender",
    palette={"male": "skyblue", "female": "pink"},
    legend=False
)

ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])

plt.show()
```



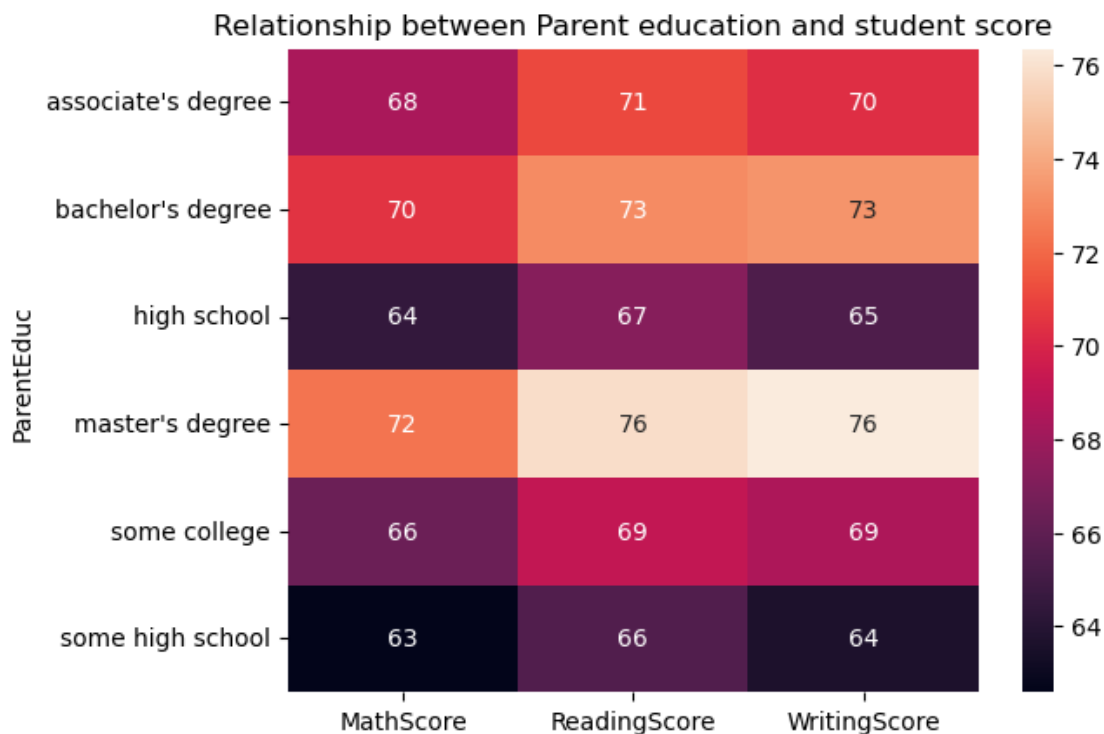
```
[188]: # From the above chart, the number of females in the data is slightly higher
↳ than the number of males.
```

1.3 Parent Education and Student Performance

```
[189]: gp0=df.groupby("ParentEduc").agg({"MathScore":'mean',"ReadingScore":
↳ 'mean',"WritingScore":'mean'})
print(gp0)
```

	MathScore	ReadingScore	WritingScore
ParentEduc			
associate's degree	68.365586	71.124324	70.299099
bachelor's degree	70.466627	73.062020	73.331069
high school	64.435731	67.213997	65.421136
master's degree	72.336134	75.832921	76.356896
some college	66.390472	69.179708	68.501432
some high school	62.584013	65.510785	63.632409

```
[190]: sns.heatmap(gp0,annot=True)
plt.title("Relationship between Parent education and student score")
plt.show()
```



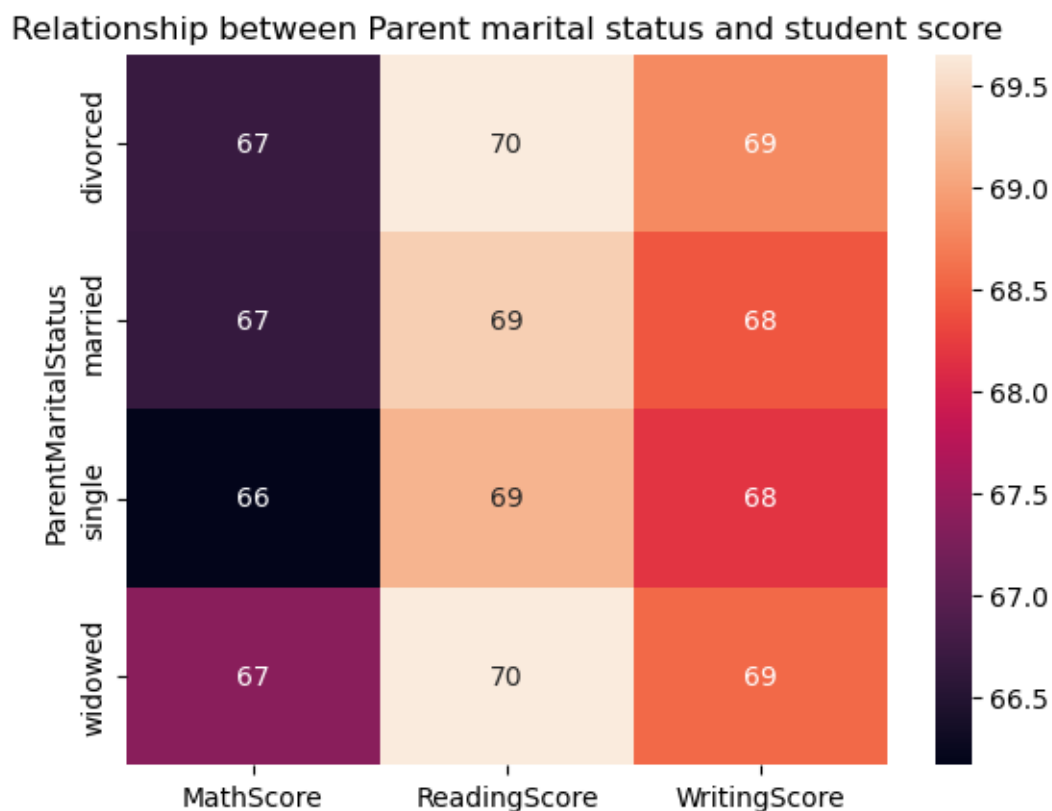
```
[191]: # From the above chart, we can see that parents' education has an impact on
        ↪ student scores.
```

```
#student whose parents have higher education levels usually score better,
        ↪ especially in reading and writing
```

```
[192]: gp1=df.groupby("ParentMaritalStatus").agg({"MathScore":'mean',"ReadingScore":
        ↪ 'mean',"WritingScore":'mean'})
        print(gp1)
```

ParentMaritalStatus	MathScore	ReadingScore	WritingScore
divorced	66.691197	69.655011	68.799146
married	66.657326	69.389575	68.420981
single	66.165704	69.157250	68.174440
widowed	67.368866	69.651438	68.563452

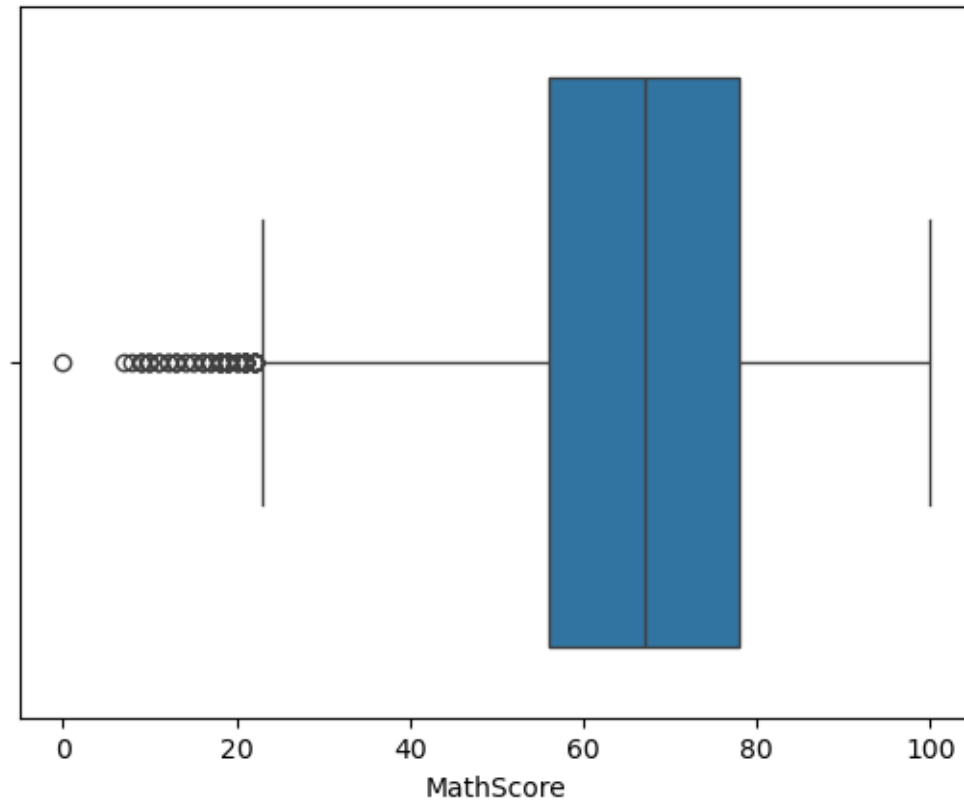
```
[193]: sns.heatmap(gp1,annot=True)
        plt.title("Relationship between Parent marital status and student score")
        plt.show()
```



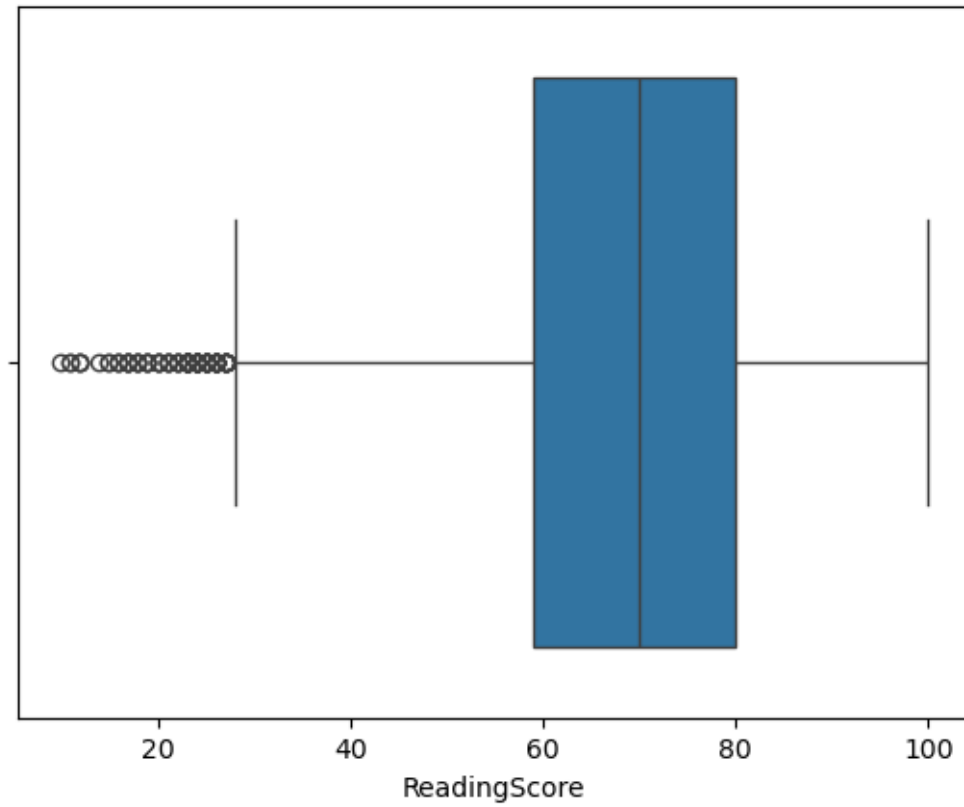
```
[194]: # Parent marital status has little or no impact on student scores.
```

1.4 Subject-wise Score Distribution

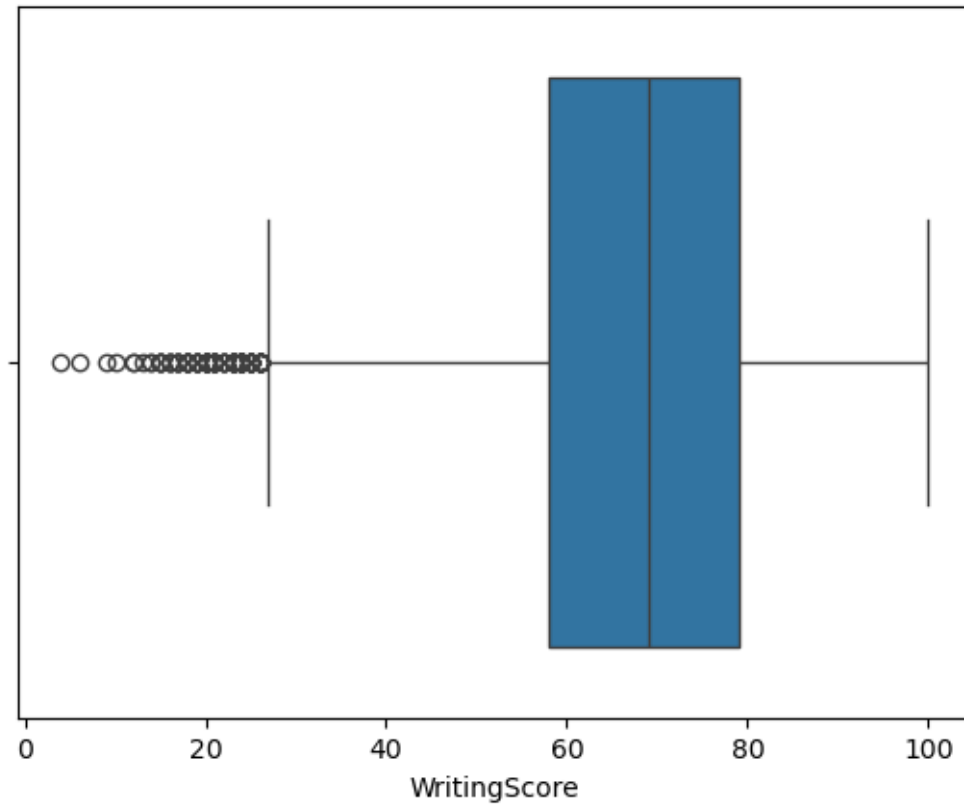
```
[195]: sns.boxplot(data=df,x="MathScore")  
plt.show()
```



```
[196]: sns.boxplot(data=df,x="ReadingScore")  
plt.show()
```



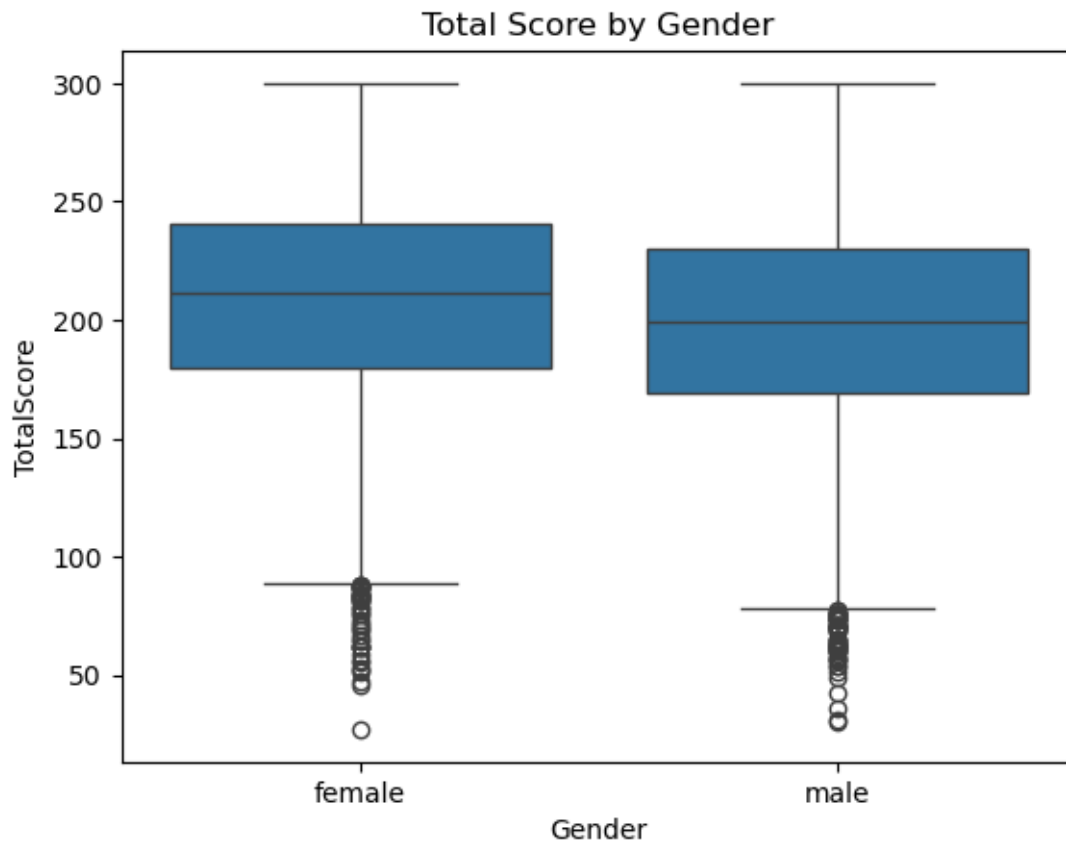
```
[197]: sns.boxplot(data=df,x="WritingScore")  
plt.show()
```

```
[198]: # Math appears to be relatively more difficult than reading and writing.
```

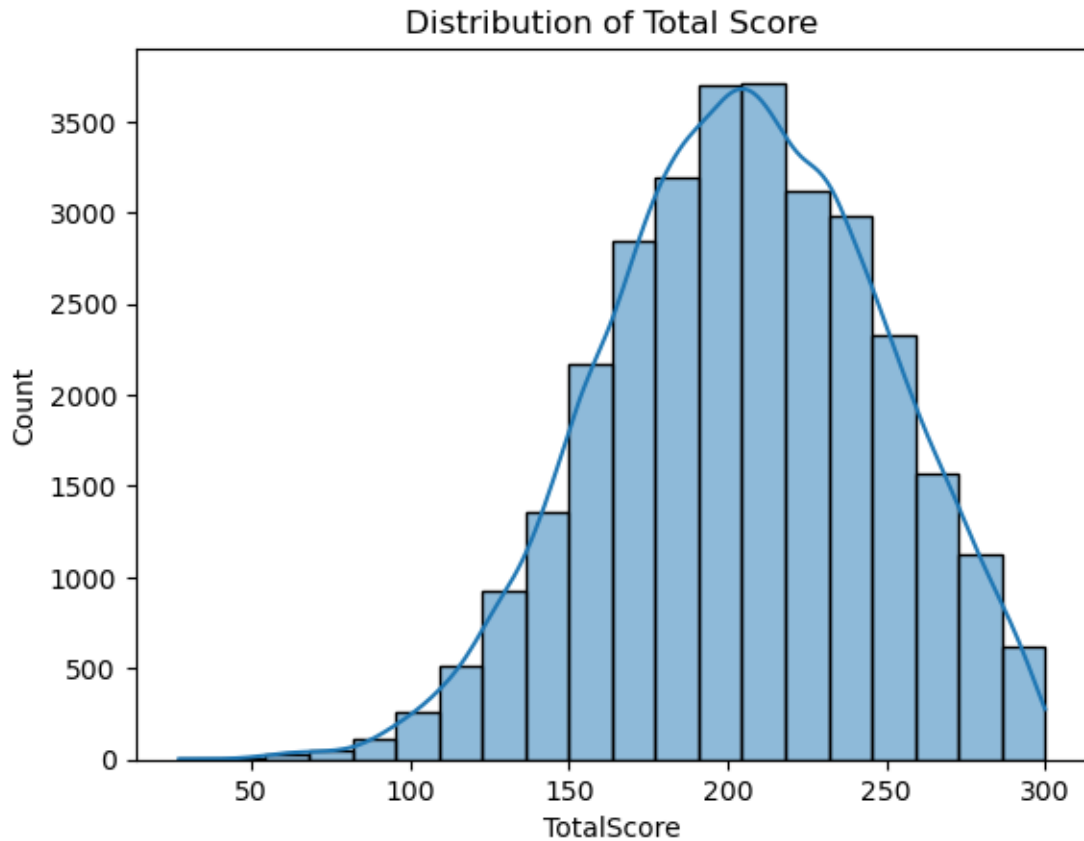
```
[199]: #total score section
df["TotalScore"] = df["MathScore"] + df["ReadingScore"] + df["WritingScore"]
```

```
[200]: sns.boxplot(data=df, x="Gender", y="TotalScore")
plt.title("Total Score by Gender")
plt.show()
```



[201]: *#The total scores of male and female students are quite close, but females have ↪
 ↪ a slightly higher median score.
 #This means both genders perform almost equally in academics.*

[202]: `sns.histplot(df["TotalScore"], bins=20, kde=True)
 plt.title("Distribution of Total Score")
 plt.show()`



[203]: *#Most students score in the middle range, with fewer students scoring extremely
 ↳ high or low.*

[204]: *#Total score gives an overall idea of student performance instead of looking at
 ↳ each subject separately*

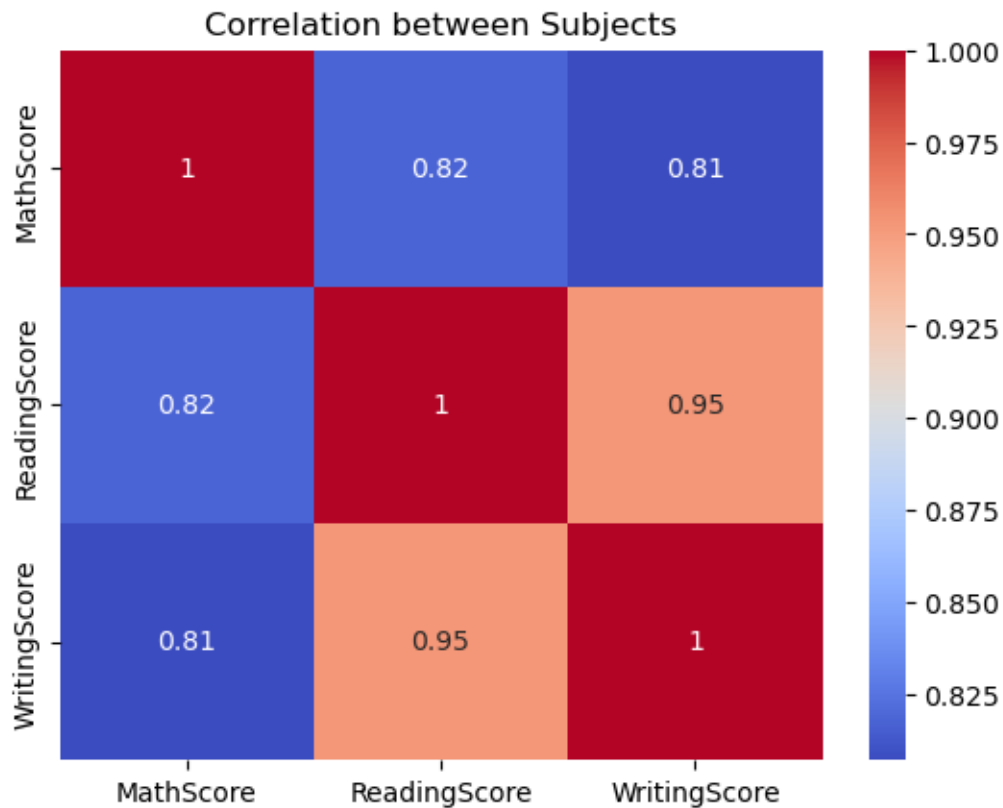
1.5 Standardized Score (Z-Score)

```
[205]: # Z-score for Total Score
df["ZScore"] = (df["TotalScore"] - df["TotalScore"].mean()) / df["TotalScore"].
↳std()
df[["TotalScore", "ZScore"]].head()
```

```
[205]:   TotalScore   ZScore
0         216  0.268557
1         247  0.983453
2         271  1.536920
3         143 -1.414906
4         229  0.568352
```

1.6 Relationship Between Subjects

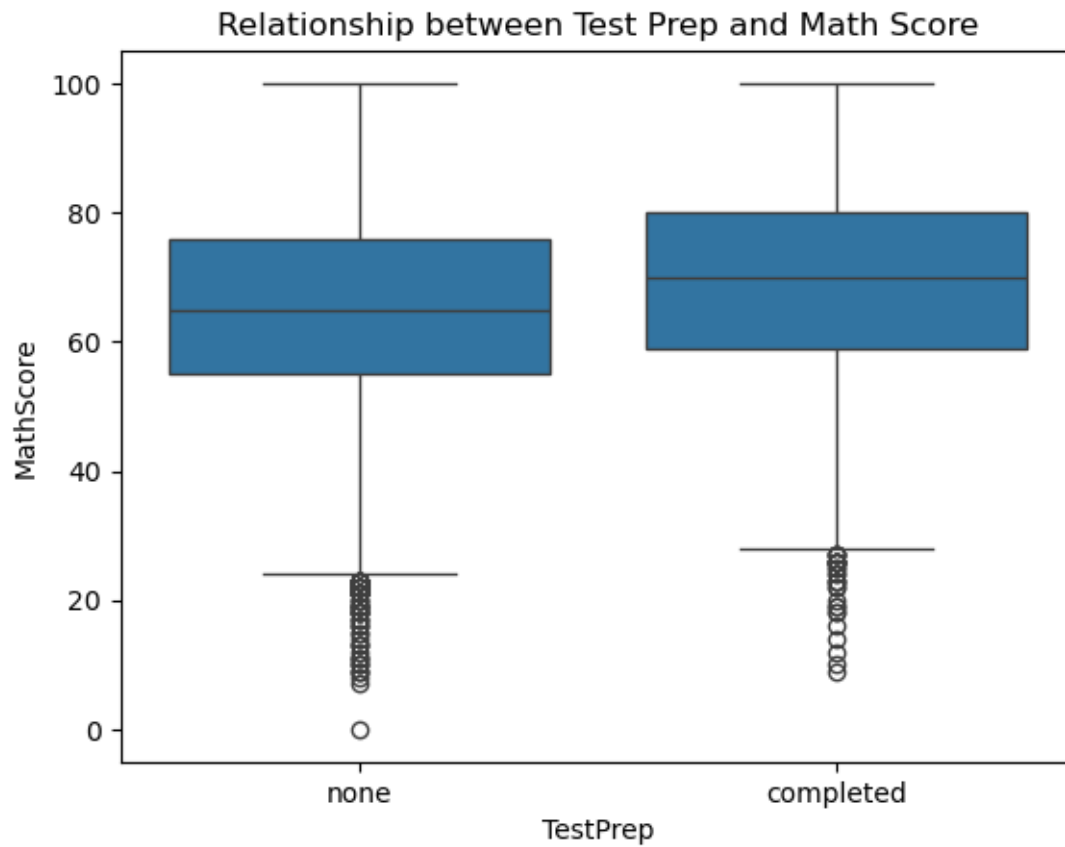
```
[206]: #Correlation Section
corr = df[["MathScore", "ReadingScore", "WritingScore"]].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation between Subjects")
plt.show()
```



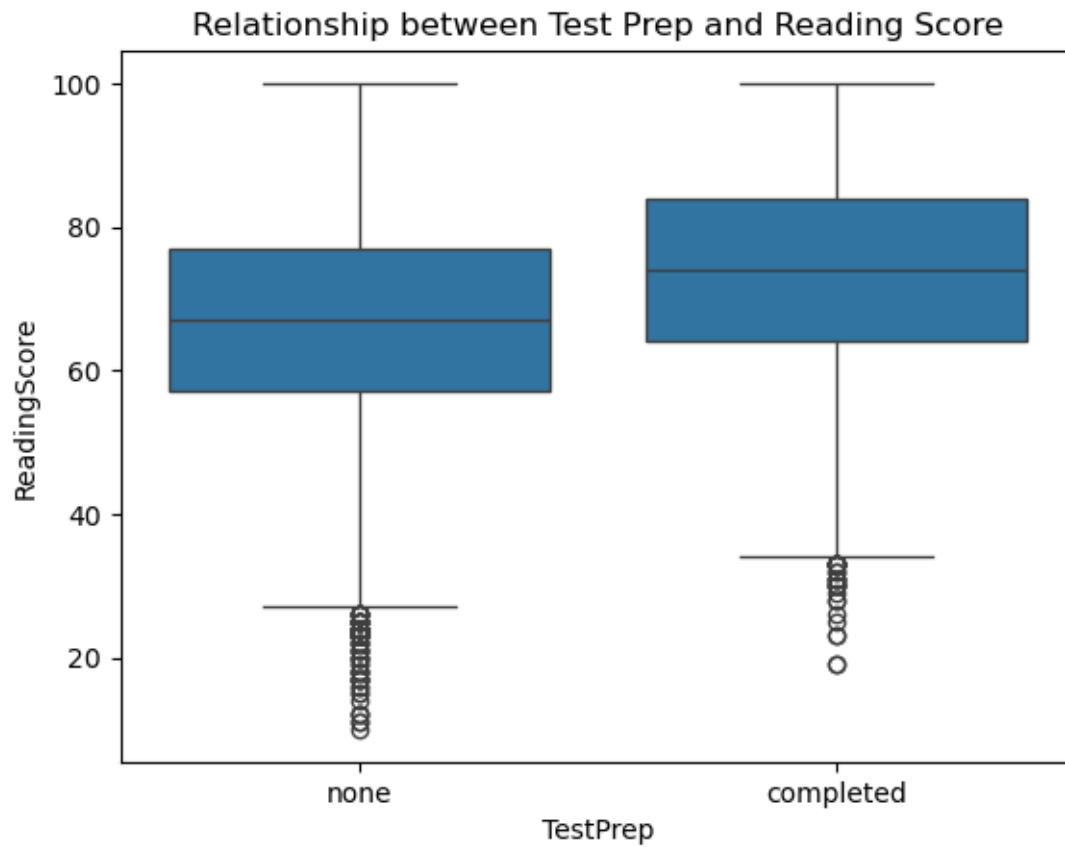
```
[207]: # Reading and writing scores are closely related, while math is slightly
        ↪different.
```

1.7 Effect of Test Preparation

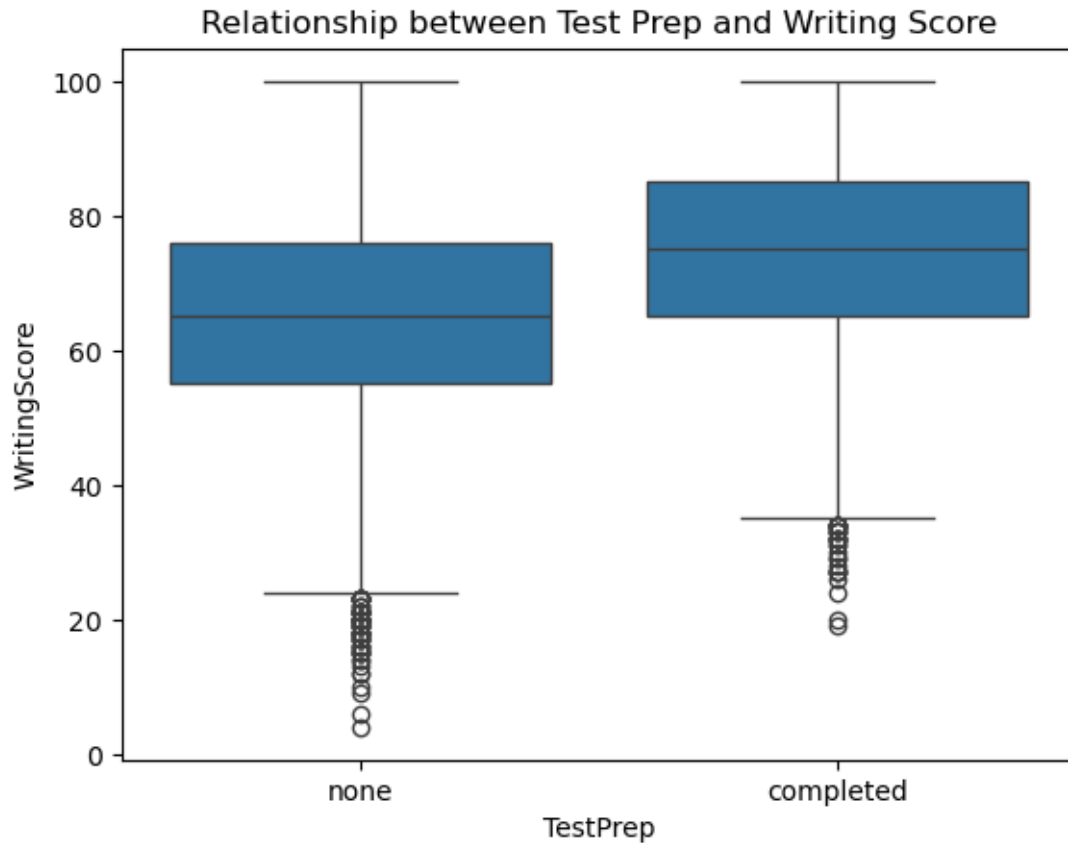
```
[208]: sns.boxplot(data=df, x="TestPrep", y="MathScore")
plt.title("Relationship between Test Prep and Math Score")
plt.show()
```



```
[209]: sns.boxplot(data=df,x="TestPrep",y="ReadingScore")  
plt.title("Relationship between Test Prep and Reading Score")  
plt.show()
```



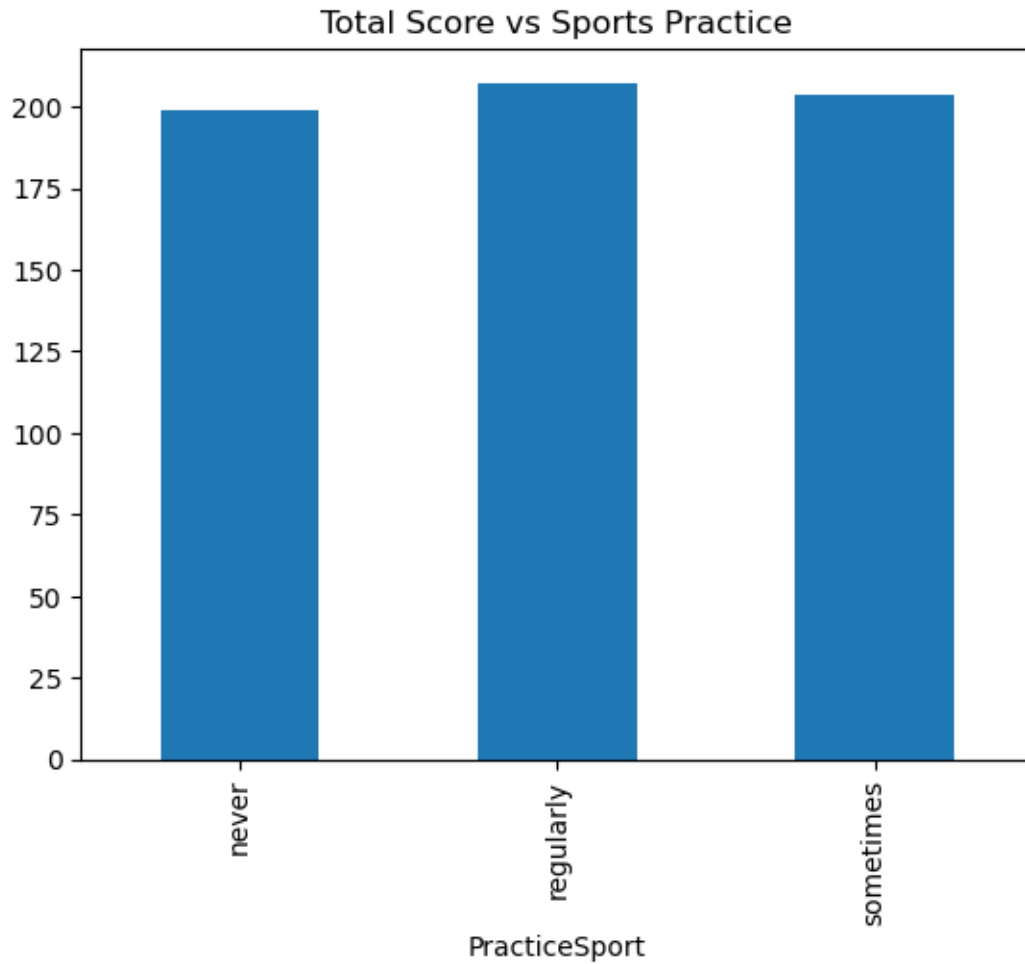
```
[210]: sns.boxplot(data=df,x="TestPrep",y="WritingScore")  
plt.title("Relationship between Test Prep and Writing Score")  
plt.show()
```



```
[211]: # Students who completed test preparation generally score higher in all three
        ↪ subjects.
```

1.8 Sports Activity and Performance

```
[212]: #sports vs performance
gp2 = df.groupby("PracticeSport")["TotalScore"].mean()
gp2.plot(kind="bar")
plt.title("Total Score vs Sports Practice")
plt.show()
```



```
[213]: #Students who practice sports regularly seem to perform slightly better.
```

```
[214]: print(df['EthnicGroup'].unique())
```

```
[nan 'group C' 'group B' 'group A' 'group D' 'group E']
```

```
[221]: #distribution of ethnic group
```

```
groupA = df.loc[(df["EthnicGroup"] == "group A")].count()
groupB = df.loc[(df["EthnicGroup"] == "group B")].count()
groupC = df.loc[(df["EthnicGroup"] == "group C")].count()
groupD = df.loc[(df["EthnicGroup"] == "group D")].count()
groupE = df.loc[(df["EthnicGroup"] == "group E")].count()
```

```
l=["group A","group B","group C","group D","group E"]
```

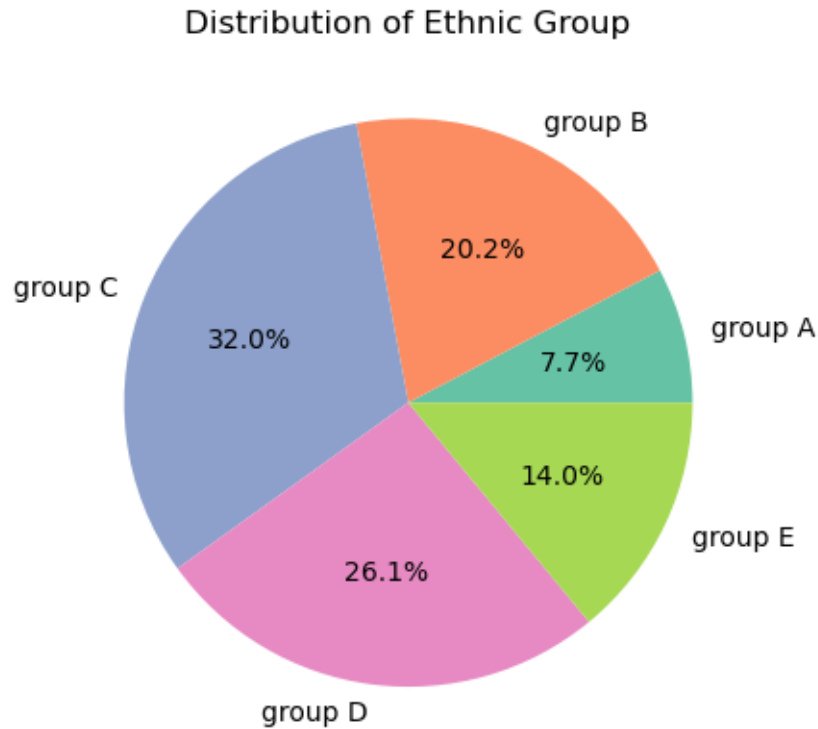
```
mlist=[groupA["EthnicGroup"],groupB["EthnicGroup"],groupC["EthnicGroup"],groupD["EthnicGroup"],
```



```

colors = ["#66c2a5", "#fc8d62", "#8da0cb", "#e78ac3", "#a6d854"]
plt.pie(mlist,labels=l,autopct='%1.1f%%',colors=colors)
plt.title("Distribution of Ethnic Group")
plt.show()

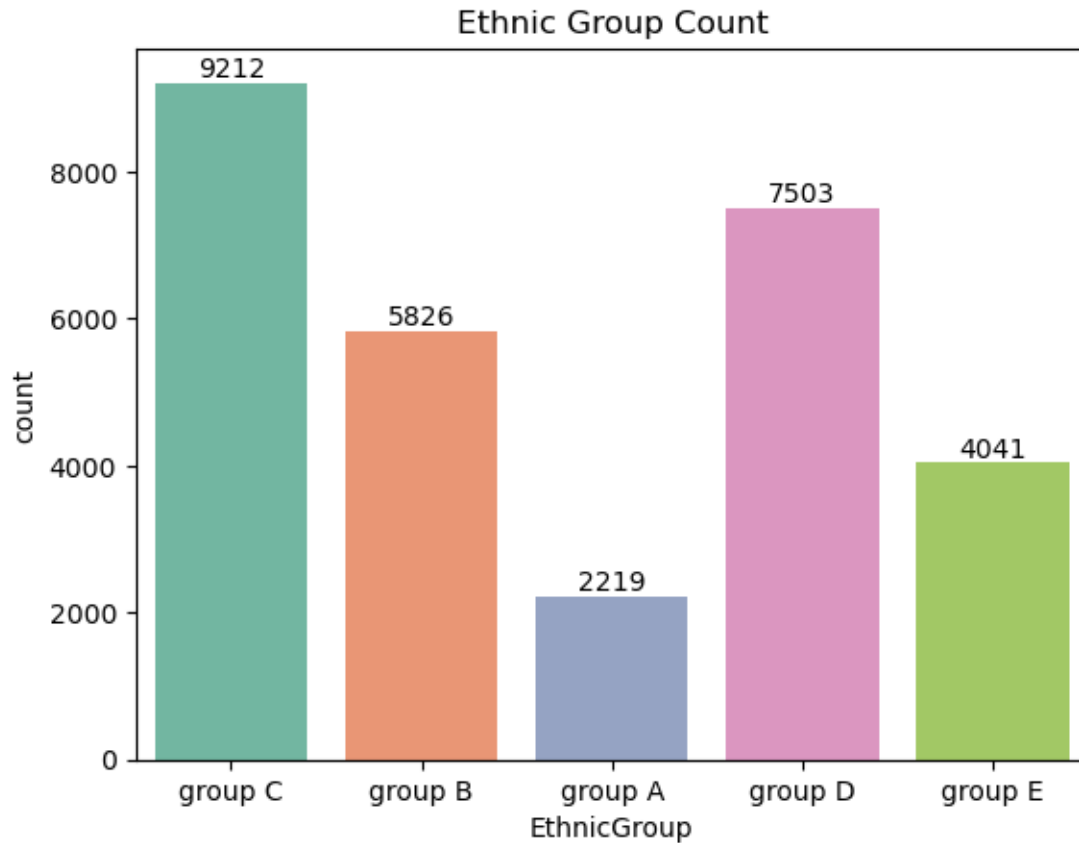
```



```

[216]: ax=sns.countplot(data=df,
    ↪x="EthnicGroup",hue="EthnicGroup",palette=colors,legend=False)
for container in ax.containers:
    ax.bar_label(container)
plt.title("Ethnic Group Count")
plt.show()

```



```
[217]: df.sort_values("TotalScore", ascending=False)[
        ↵
        ↪["Gender", "EthnicGroup", "MathScore", "ReadingScore", "WritingScore", "TotalScore"]
    ].head(5)
```

```
[217]:
```

	Gender	EthnicGroup	MathScore	ReadingScore	WritingScore	TotalScore
2484	female	group A	100	100	100	300
20396	female	group A	100	100	100	300
2325	male	group E	100	100	100	300
8883	male	group D	100	100	100	300
20498	female	group D	100	100	100	300

```
[218]: df.sort_values("TotalScore")[
        ↵
        ↪["Gender", "EthnicGroup", "MathScore", "ReadingScore", "WritingScore", "TotalScore"]
    ].head(5)
```

```
[218]:
```

	Gender	EthnicGroup	MathScore	ReadingScore	WritingScore	TotalScore
55	female	group C	0	17	10	27
14494	male	group D	7	11	12	30

18385	male	group B	10	12	9	31
4719	male	group B	19	11	6	36
30389	male	group B	24	14	4	42

[219]: *#Top and bottom scoring students to understand the performance range.*

1.9 Conclusion

[220]: *#Overall, this project shows how factors like test preparation, parents' ↪ education, study habits, and activities can influence student performance.*